Roy Yao

949-665-9497 • roy_h@berkeley.edu • https://www.linkedin.com/in/27yaoroy/ • https://royyaoyh.github.io/

EDUCATION:

University of California, Berkeley **(Aug 2023- May 2027)**

B.S. Electrical Engineering & Computer Science, Bioengineering, GPA: 3.84/4.00

Honors & Awards: Dean's List, EECS Honors Program, Bioengineering Honor Society, Rose Hills Summer Scholarship, GLOBE Discovery Scholarship. **Relevant Coursework:** Machine Learning, Computer Vision, Operating Systems, Signals & Systems, Robotics, Optimization, Algorithms, MRI, Medical Imaging, Bioinstrumentation, BioMEMS/Medical Devices

EXPERIENCES:

Imaging Scientist Intern, *Varex Imaging*, San Jose **(Jun 2026 - Aug 2026)**

- Co-authored a submission to SPIE Medical Imaging 2027 on charge-sharing correction for photon-counting detectors; developed Geant4 Monte Carlo simulations and multi-material phantoms to validate detector response against measured data. Demonstrated a 2x increase in SNR for the novel charge-sharing correction mode in simulation and measured data.
- Debugged charge-sharing correction algorithms, developed an ImageJ analysis plugin, and standardized calibration workflows to improve quantitative analysis and reproducibility.

Undergraduate Researcher, *Conolly Lab*, UC Berkeley **(Aug 2025 - Present)**

- Designed and simulated a low-noise analog front end for magnetic particle imaging (MPI) in LTspice; implemented the preamplifier PCB in KiCad, optimized noise matching, and evaluated noise performance across a 5 MHz bandwidth.
- Hand-soldered, debugged, and characterized circuit performance using bench instrumentation, including oscilloscopes, function generators, network analyzers, and digital multimeters.

Summer Research Intern, *Chen Lab*, National Tsing Hua University **(Jun 2025 - Jan 2026)**

- Developed Python/OpenCV video-analysis and signal-processing pipelines to detect nanoparticle transit events, using filtering, temporal alignment, and correlation against lock-in amplifier measurements. Enabled video-based particle counting independent of photodiode DAQ by validating detected events against experimental measurements.

Undergraduate Researcher, *Ganguly Lab*, UC San Francisco **(Oct 2023 - May 2024)**

- Applied empirical mode decomposition (EMD) and channel-dropout data augmentation to brain–computer interface (BCI) signals; implemented MLP, CNN, RNN, and LSTM models for neural signal-based action classification.

PROJECTS:

Robotics Manipulation & Mobile Navigation, UC Berkeley **(Aug 2026 - Present)**

- **UR7e Robotic Arm**: Developed a ROS 2 manipulation pipeline using Intel RealSense point-cloud perception, MoveIt inverse kinematics, and trajectory planning for a 6-DOF robotic arm.
- **TurtleBot 3 Autonomous Navigation**: Implemented LiDAR-based mapping and probabilistic occupancy-grid updates in ROS 2 for autonomous navigation and motion planning.

Microfluidic Single-Cell Dispensing Device, UC Berkeley **(Apr 2026 - May 2026)**

- Designed and tested a passive hydrodynamic trapping device for 40-µm polyacrylamide bead cell analogs in a 9-mm-pitch, 96-well-compatible layout; achieved 81.25% single-bead trapping (13/16) with 80-µm channels.

EMG-Based Word Detection Sensor, UC Berkeley **(Apr 2025 - May 2025)**

- Built a facial EMG acquisition system using Ag/AgCl electrodes, dual AD620 instrumentation amplifiers, and an Arduino MKR Zero; measured 4,036× gain and a 21–495 Hz bandwidth, and implemented autocorrelation-, similarity-, and CNN-based word classification. Achieved ~80% accuracy in word recognition.

Silent Speech Decoding, Neurotech@Berkeley, UC Berkeley **(Aug 2025 - Dec 2025)**

- Analyzed synchronization and extracted features across audio, lip landmarks, RGB/depth video, and laser-speckle data; trained a CNN on augmented speckle time series, achieving 71.15% accuracy across 30 classes.

Pintos Operating System, UC Berkeley **(Jan 2026 - May 2026)**

- Implemented process management, system calls, thread synchronization, priority scheduling, and file-system operations; debugged kernel and concurrency issues using GDB.

SKILLS:

- **Programming:** Python, C++, C, MATLAB, Java, SQL
- **ML & Signal Processing:** PyTorch, NumPy, OpenCV, Computer Vision, Digital Signal Processing
- **Hardware & Simulation:** LTspice, KiCad, Bench Instrumentation, Arduino, Circuit Design, PCB Design, COMSOL
- **Robotics:** ROS 2, MoveIt, Intel RealSense, LiDAR Mapping
- **Additional Tools:** Git, SolidWorks, ImageJ, Geant4